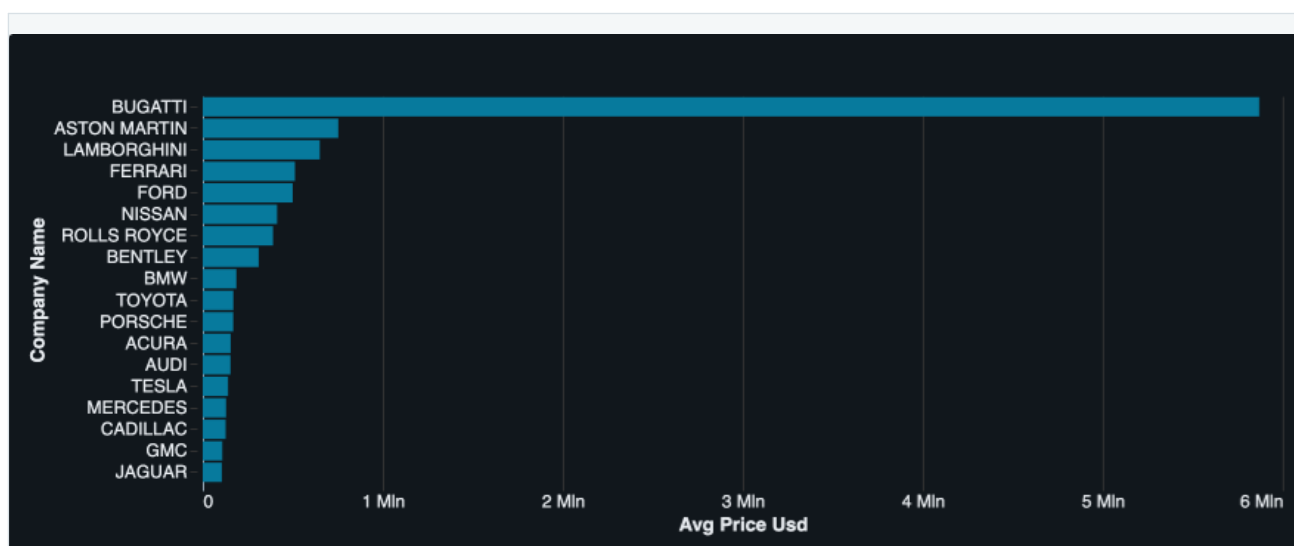


Luxury Brand Pricing — When Price Leaves Performance Behind

Companion note to the data-cleaning notebook · Gold table brand_metrics_luxury



THE INSIGHT

The point isn't that Bugatti is the most expensive brand — in the wider market other marques run far higher, and these are simply the per-brand averages this dataset produced. The point is what happens when you put price next to performance. Bugatti is also the **most powerful** car in the set (~1,850 hp), yet its price still works out to roughly **\$3,000 per horsepower** ($\sim \$5.9\text{M} \div \sim 1,850\text{ hp}$) — about **4× the Luxury segment's \$714/hp** and ~20× an Economy car's \$163/hp. At this end of the market, price has decoupled from performance entirely: you're paying for the brand and the prestige of owning one, not for the horsepower.

What this chart is

Average selling price per brand, within the Luxury price tier (petrol, showroom, light vehicles). Each bar is **one brand's own mean** — not a pooled segment average — so a tall bar reflects that brand's line-up in this dataset, nothing more.

Why it ships with the cleaning notebook

This brand view is where the dataset's **extreme values surface**, so it doubles as the visual companion to the cleaning work. Bugatti's ~\$5.9M is a genuine car and stays; a ~\$5M "Mazda," by contrast, turned out to be a race prototype and was excluded via the `is_showroom_model` flag. The rule throughout — **exclude the impossible, keep the merely surprising** — is what makes the price-per-hp figures above trustworthy rather than artefacts of dirty data.

Built on Databricks (PySpark · Spark SQL · Delta Lake · Unity Catalog · AI/BI). Price-per-hp reference values from the Gold table `price_per_hp_petrol`. The ~\$3,000/hp figure is an illustrative price ÷ peak-hp calculation.